

odically for plotting the same on Fleetview Mapping software, and generating exception reports for analysing driving conditions of a vehicle.

FleetView's vehicle management services module helps predict what is going to happen, rather than what has happened. This empowers a company to make strategic decisions to protect its fleet. Mobile Eye helps enhance fleet operations by providing real-time actionable intelligence to take charge of the supply chain. It helps generate data from the most relevant sources of a journey and converts it into strategic decisions for the business. A good fleet and supply chain management solution provides a clear performance standard for customers of the business, giving a clear competitive edge by improved visibility across operations and consistent, dependable performance.

Dr Sebastian Ritz, founder and chief executive officer, iNNOVO Cloud, says, "We provide a solution package for all scenarios, from leasing a fully-configured container to a turnkey container data centre, including standardised private cloud as a managed or self-managed service. Components can be configured as the smallest hardware unit in the customer's on-site data centre or in a virtualised environment. These blobs (binary large objects) provide IT output for specific requirements such as high-performance computing, big data systems, high input/output operations per second and database systems with high demand for main memory."

In the installation offered by Rittal and iNNOVO Cloud, blobs of this kind consist of two IT racks (including power supplies): cooling and standardised active IT components. HPC blobs can be incorporated for high-performance computing applications. Big data blobs are used if a lot of storage and RAM are needed. These are provided locally or as a cloud service. The entire system is also available as a container solution

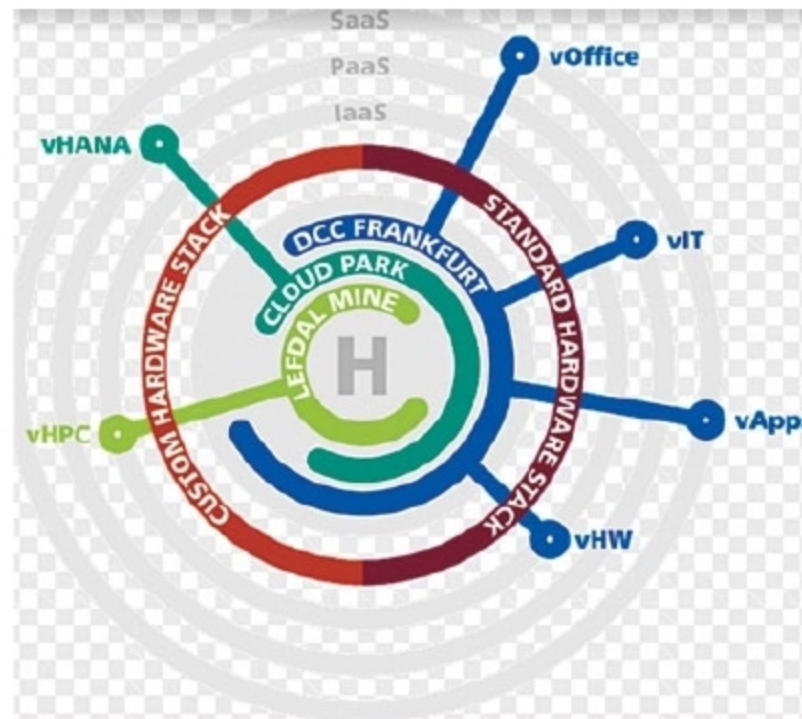


Fig. 5: Data centres migrating step-by-step to cloud (Credit: Rittal GmbH & Co. KG)

for optimum mobility and security. Once all components are linked, the data centre itself can be offered as a virtual service.

Cloud federation benefits and implementation

Federation of cloud resources (FCR) is facilitated through network gateways to create a hybrid cloud computing environment that connects public, private and/or community clouds. FCR allows a client to choose the best cloud services provider to meet a particular business or technological need, in terms of flexibility, cost and availability of services. It allows an enterprise to implement innovative security models for user access and distribute workloads between disparate networks.

FCR can be implemented behind a firewall to provide a list of cloud services to clients. Cloud providers grant permission to clients to specify an addressing scheme for each server. This enables users to access cloud services without the need for reconfiguration when using resources from different service providers.

Benefits of IaaS

According to Javier Barabas, IBM cloud and cognitive technical leader, cloud computing offers IT managers solutions for more data storage, greater and quicker software accessibility, and cost and effort optimisation. Depending on the needs of the client,

IaaS can deliver the following benefits:

1. It enables virtualisation of administrative tasks, freeing up time for other work.

2. The cloud computing service provider manages the infrastructure, saving time and cost.

3. Better resources mean more and constant processing power. A particular resource can be rented as needed.

4. Infrastructure is scalable depending on processing and storage needs.

5. It avoids the expense and complexity of buying and managing own physical servers and data centre infrastructure.

6. The service can be accessed from anywhere using the Internet, available through a public or private cloud.

7. If one aspect of the service fails, the service remains unaffected, as data is stored on the cloud. So, there are no single points of failure.

8. It can reduce cost, and access applications and data during a disaster or outage.

9. It enables quick scaling up of resources to accommodate spikes in demand for applications during the holidays. Scaling down can be done when activity goes down to save money.

10. A cloud service provider provides security for applications and data.

An increasing number of companies are obtaining cloud-based parts for their IT infrastructure. Costs of purchasing, setting up and operating the planned IT environment play a big role in this decision.

More important are the strategic opportunities that have opened up by the cloud. For example, IT organisations can provide a standards-based IT infrastructure, while also responding quickly and flexibly to constantly growing resource requirements. IT as a Service (ITaaS) operating model has emerged in recent years, with a focus on business services. Design scope for services of companies opting for ITaaS depends on cloud providers. **EFY**